msvtx_nTGCExtendedStation

overview · all regions and selections · rank 6 / 169

rank 6 / 169 by endcap / ht40 / D_equal

region	selection	D_equal	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg
endcap	all	0.700	0.766	0.599	0.585	0.681	0.774	0.799	0.828	34,970
endcap	ht40	0.683	0.753	0.580	0.567	0.662	0.761	0.788	0.819	7,390
endcap	bkg split	0.040	below 27,580 / above 7,390							
barrel	all	0.034	0.041	0.072	0.060	0.033	0.033	0.056	0.060	1,089
barrel	ht40	0.030	0.041	0.055	0.060	0.029	0.024	0.054	0.057	220
barrel	bkg split	0.027	below 869 / above 220							
all	all	0.448	0.490	0.402	0.351	0.448	0.519	0.494	0.519	36,059
all	ht40	0.436	0.482	0.384	0.334	0.431	0.509	0.488	0.514	7,610
all	bkg split	0.039	below 28,449 / above 7,610							

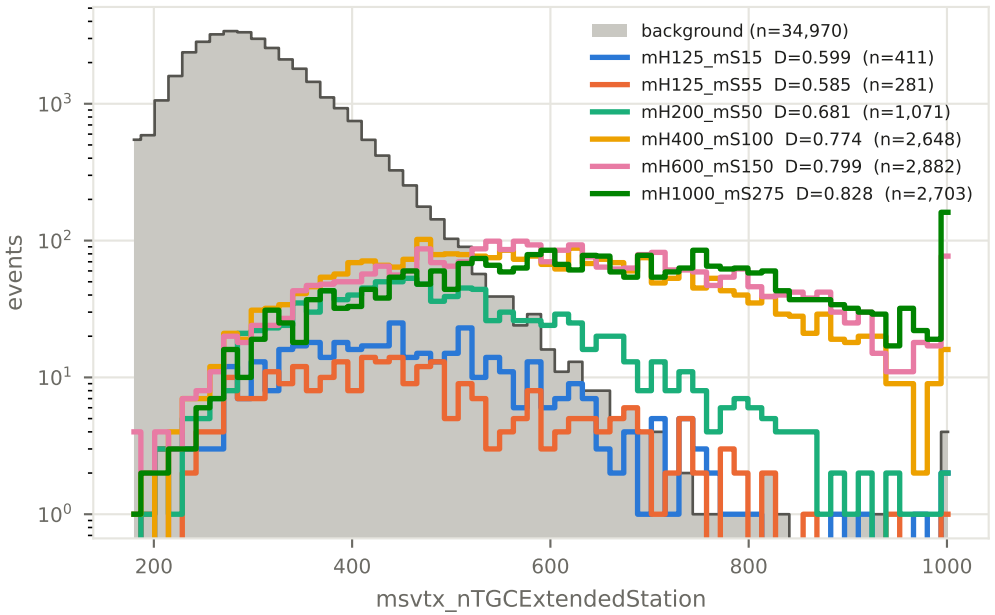
selections: all = every event; ht40 = htmissScalar_NOSYS >= 40 GeV, signal and background
pages: per region -> [all] 2x2, [ht40] 2x2, [background below vs above] 1x2

msvtx_nTGCExtendedStation

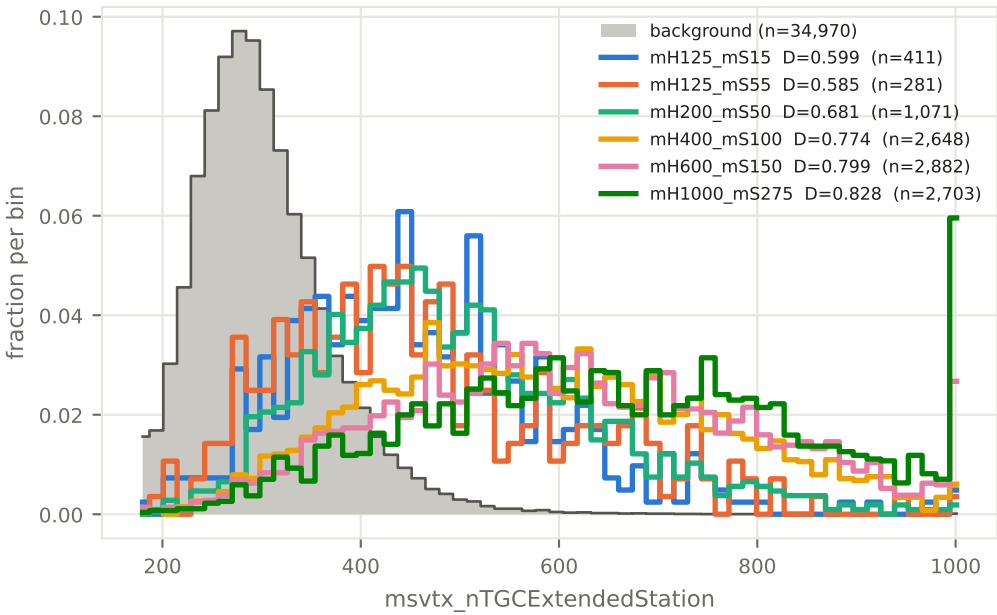
region: ENDCAP · selection: all (every event) · rank 6 / 169

D_equal 0.700 D_pooled 0.766 min 0.585 (mH125_mS55) max 0.828 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

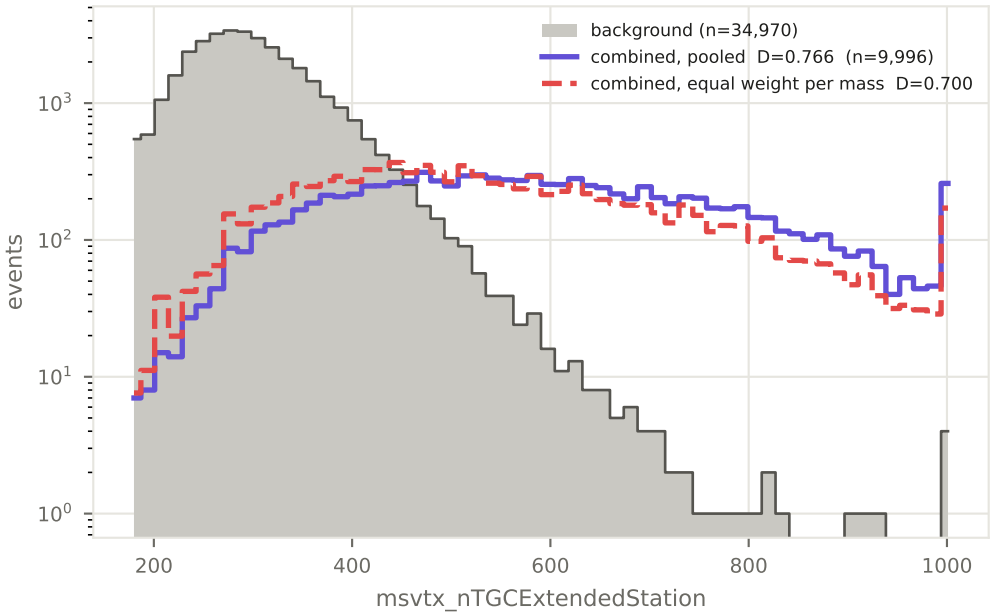
each mass point vs background · raw



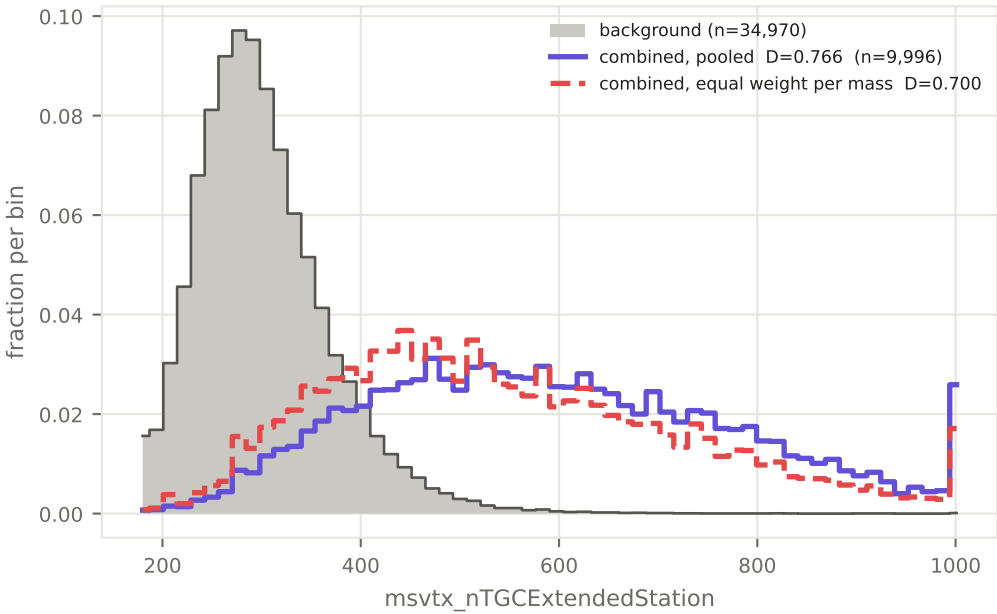
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



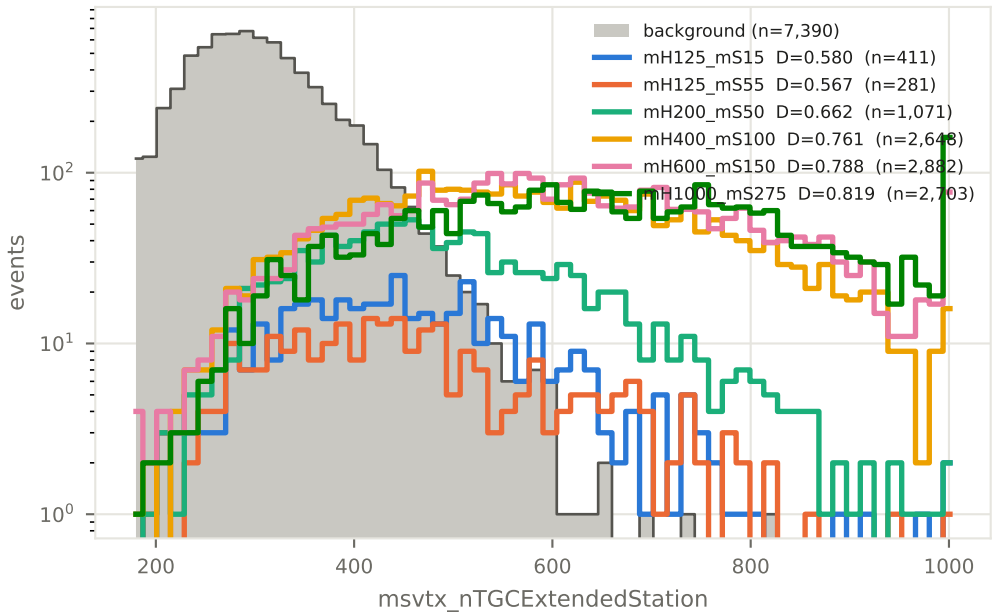
edge bins include under/overflow · NaN and -999 removed

msvtx_nTGCExtendedStation

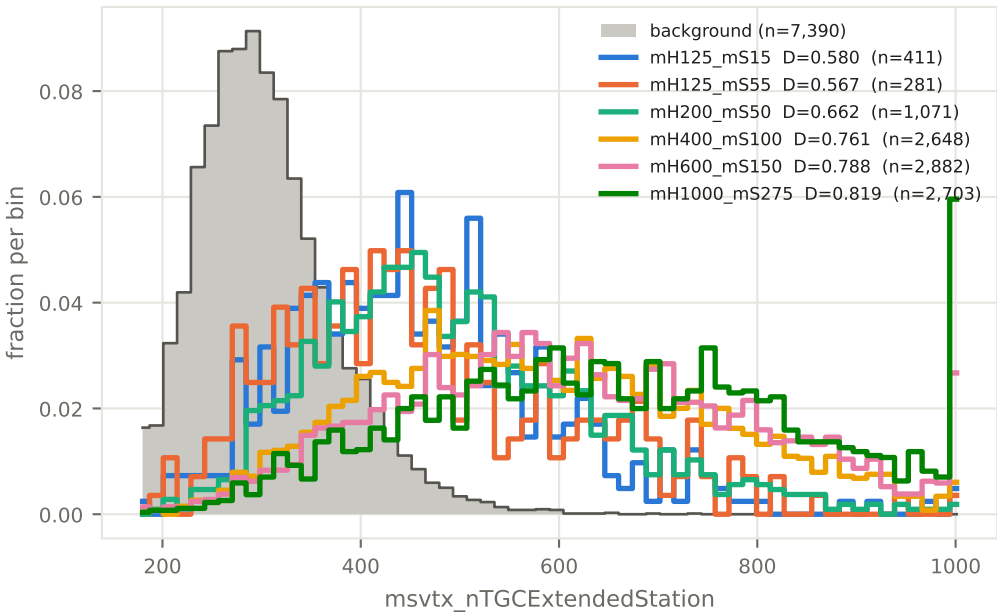
region: ENDCAP · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 6 / 169

D_equal 0.683 D_pooled 0.753 min 0.567 (mH125_mS55) max 0.819 (mH1000_mS275) bkg n = 7,390 sig n = 9,996

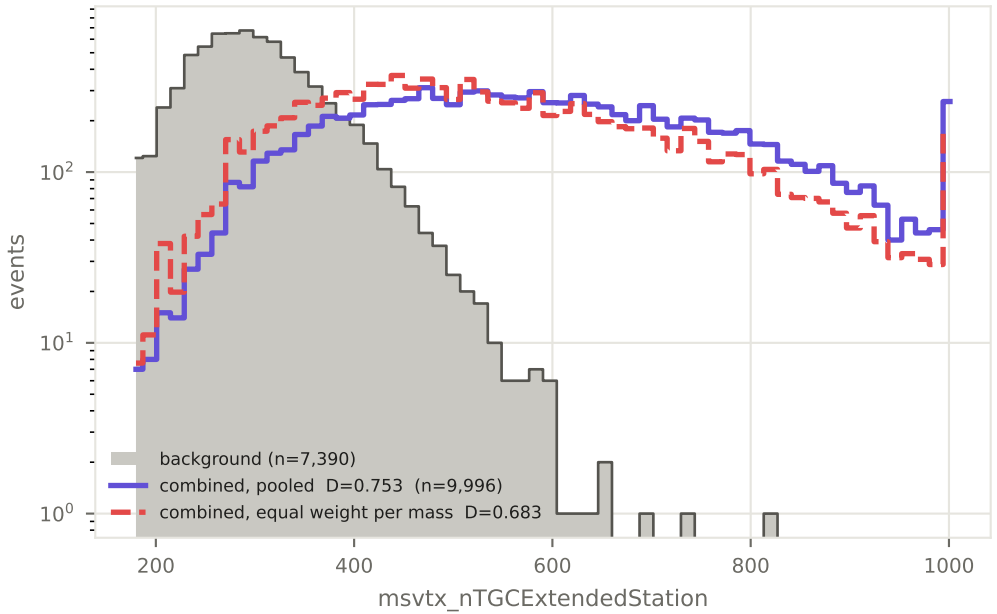
each mass point vs background · raw



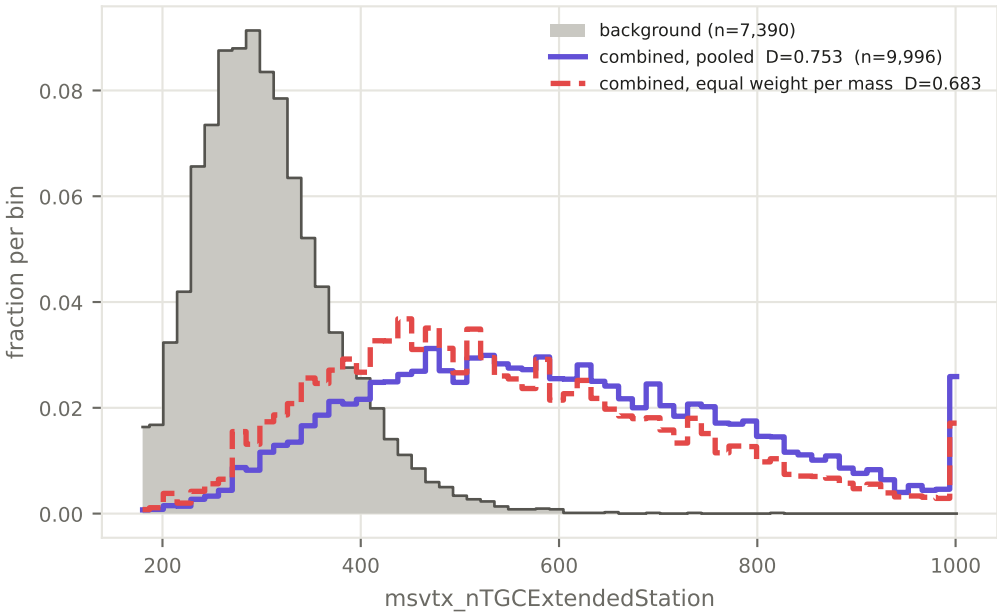
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized

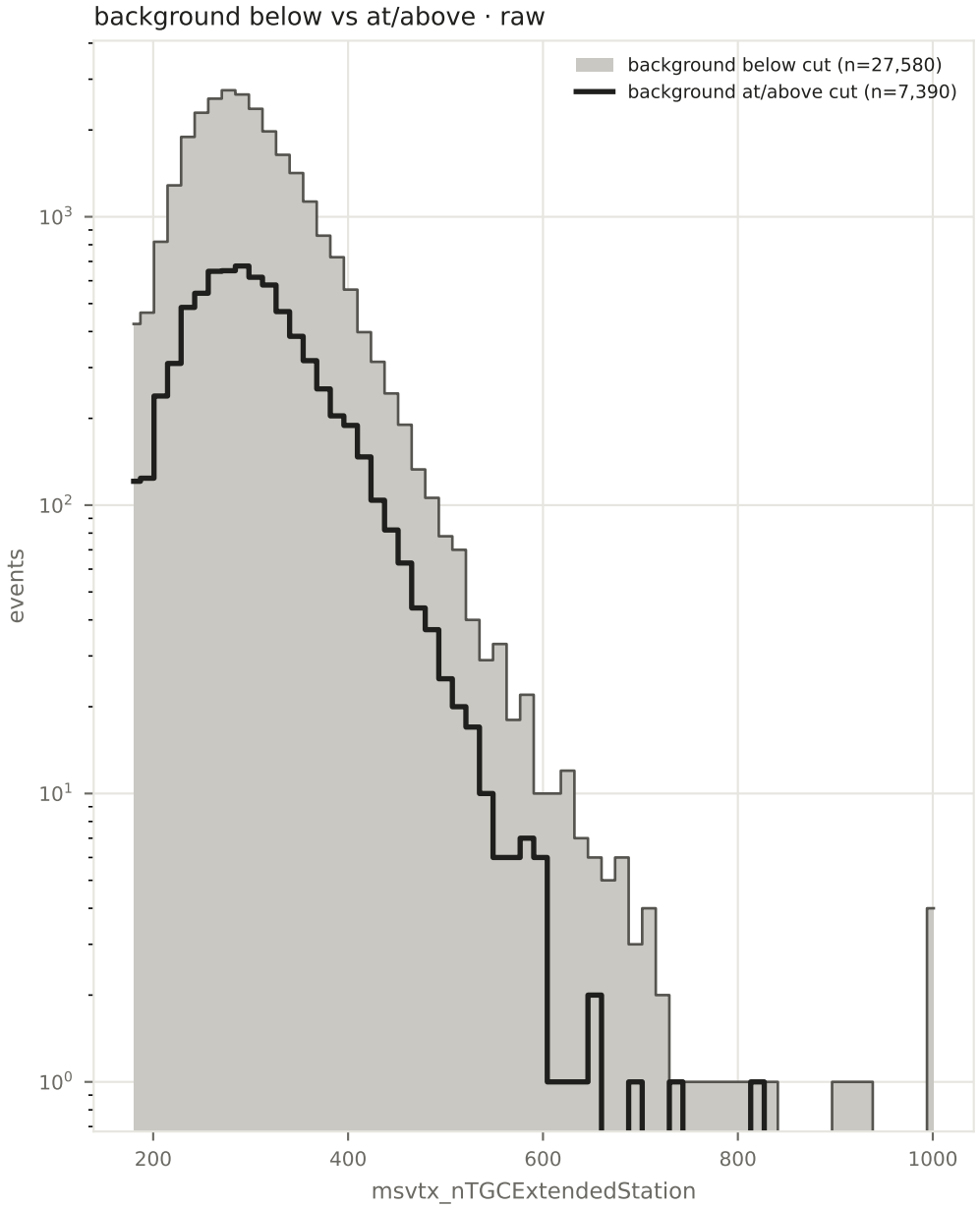


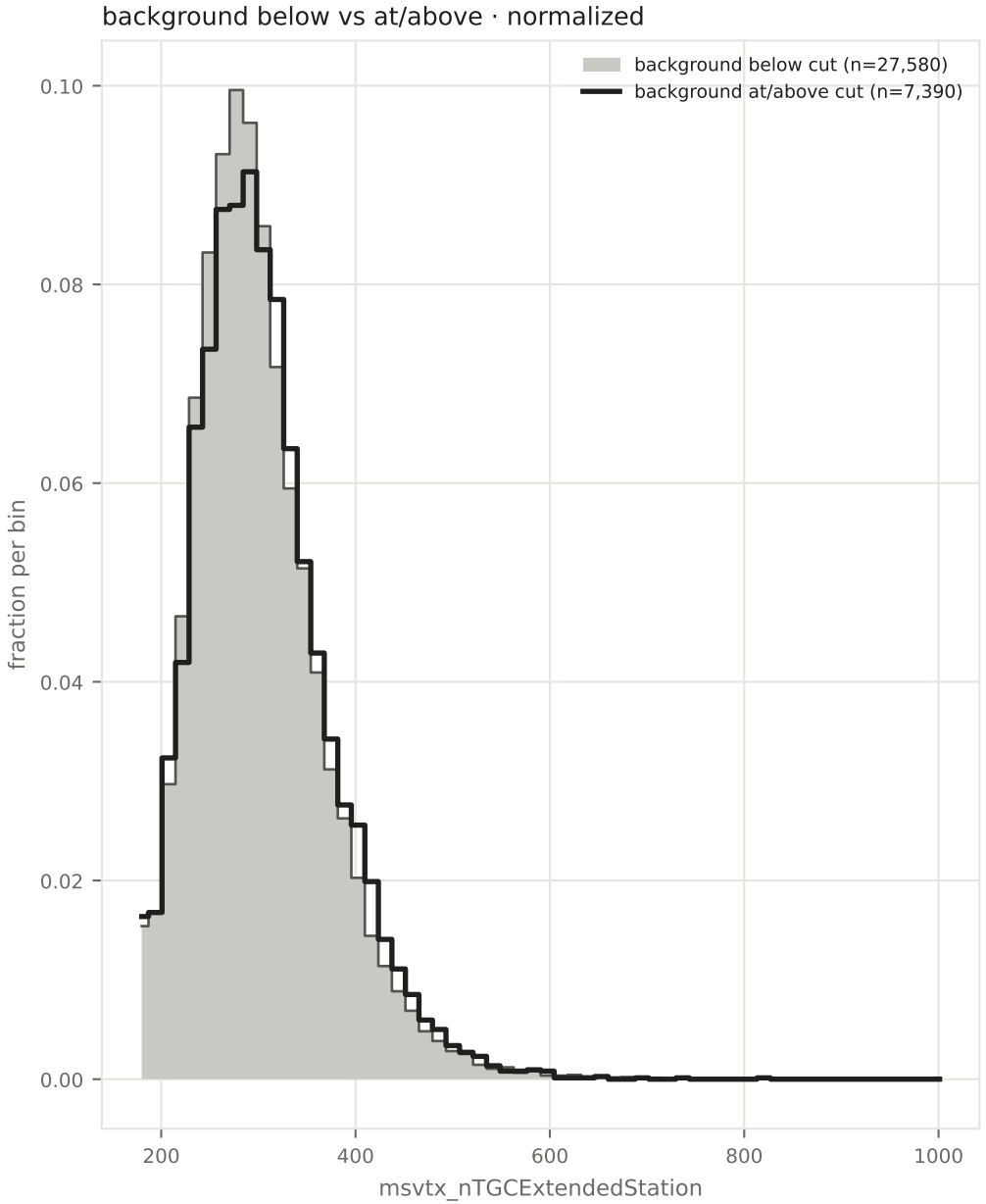
edge bins include under/overflow · NaN and -999 removed

msvtx_nTGCExtendedStation

region: ENDCAP · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 6 / 169

D(below vs above) 0.040 n below = 27,580 n above = 7,390 large D = background shape changes across the cut



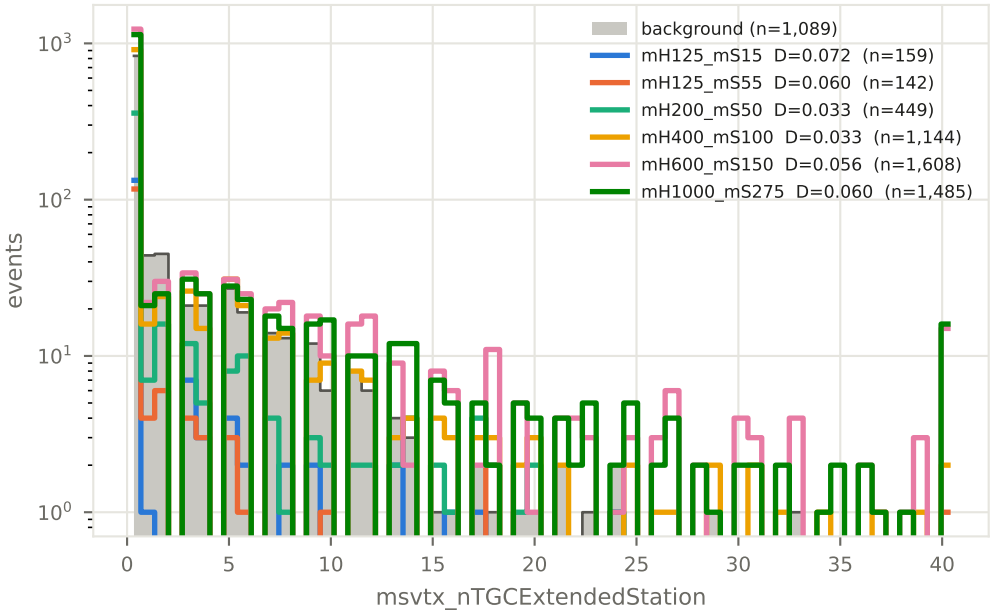


msvtx_nTGCExtendedStation

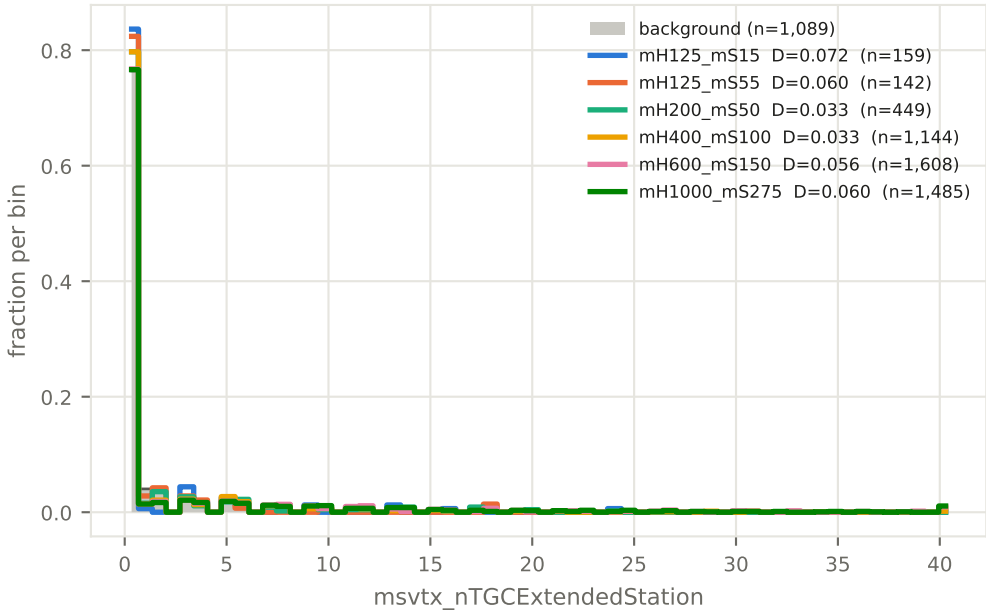
region: BARREL · selection: all (every event) · rank 6 / 169

D_equal 0.034 D_pooled 0.041 min 0.033 (mH400_mS100) max 0.072 (mH125_mS15) bkg n = 1,089 sig n = 4,987

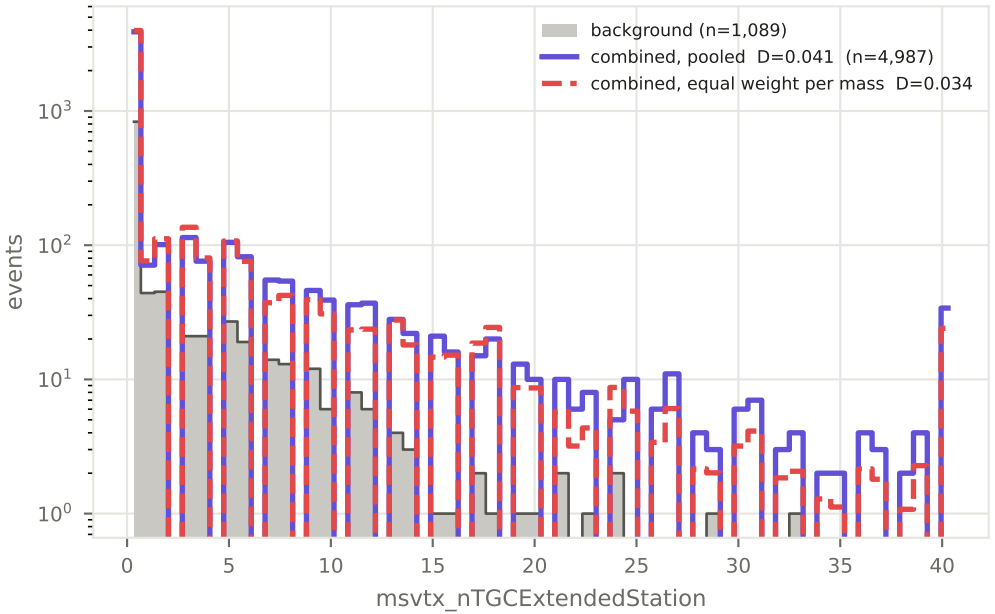
each mass point vs background · raw



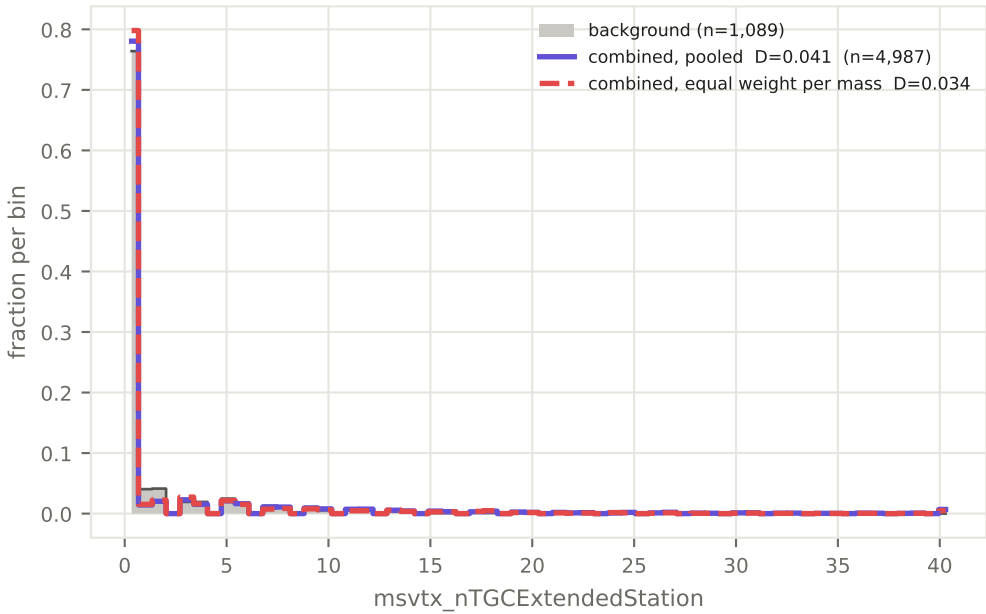
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



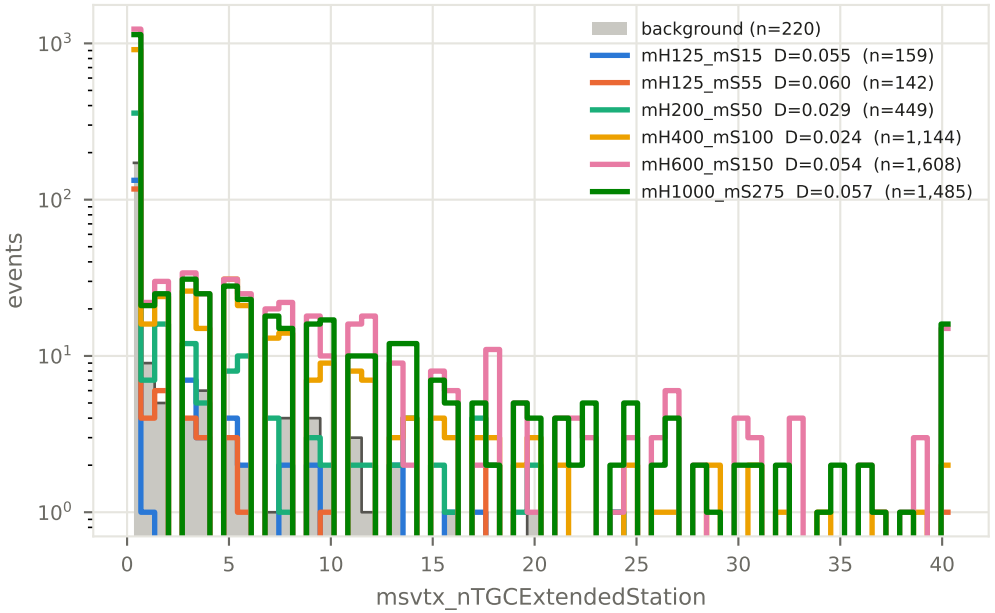
edge bins include under/overflow · NaN and -999 removed

msvtx_nTGCExtendedStation

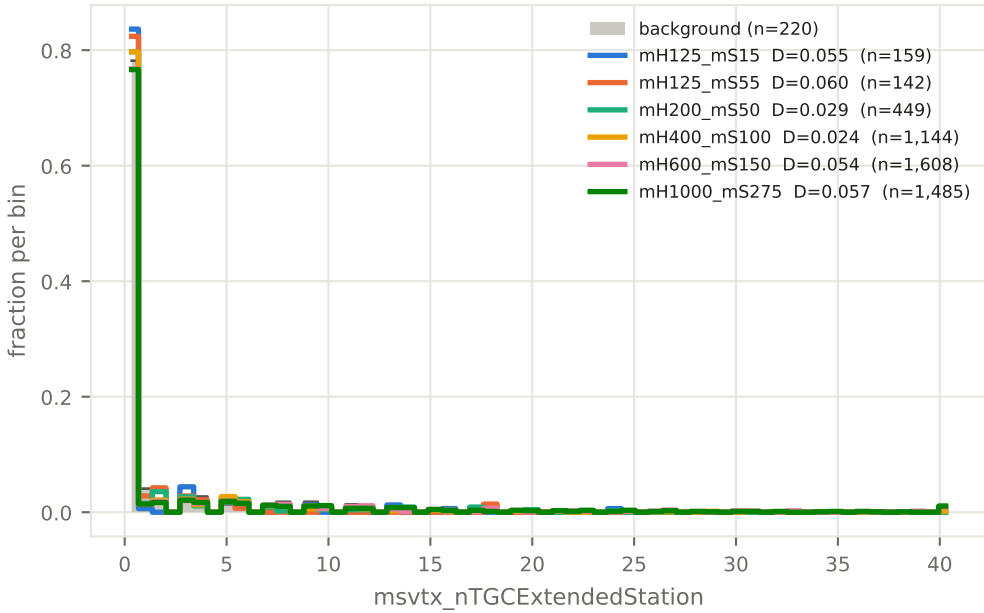
region: BARREL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 6 / 169

D_equal 0.030 D_pooled 0.041 min 0.024 (mH400_mS100) max 0.060 (mH125_mS55) bkg n = 220 sig n = 4,987

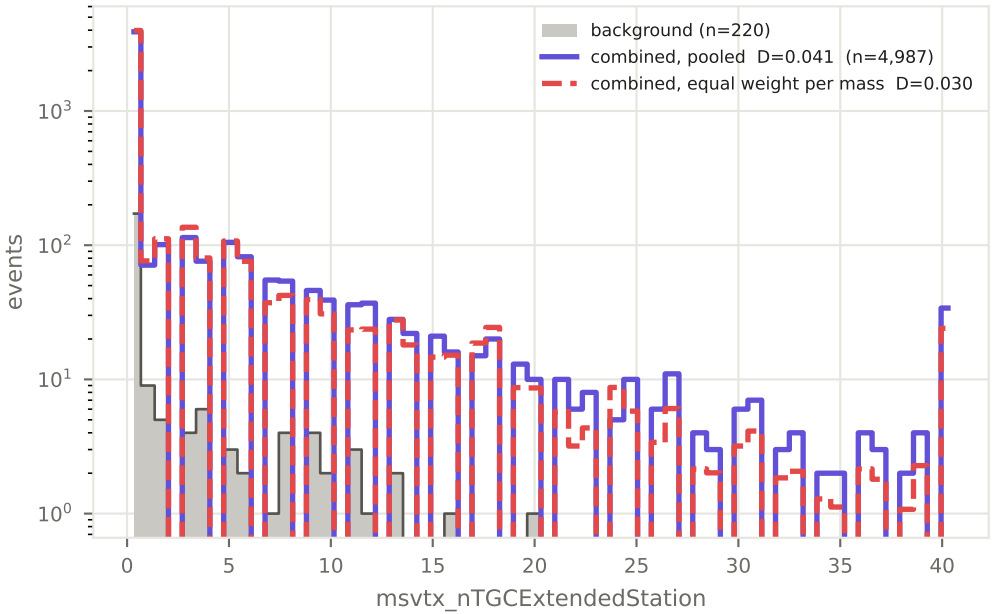
each mass point vs background · raw



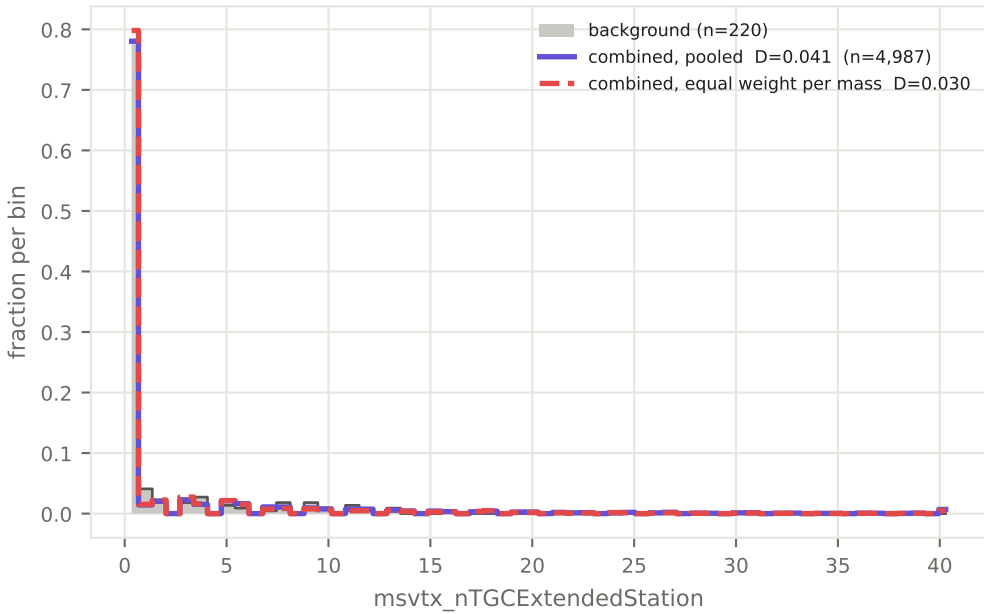
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized

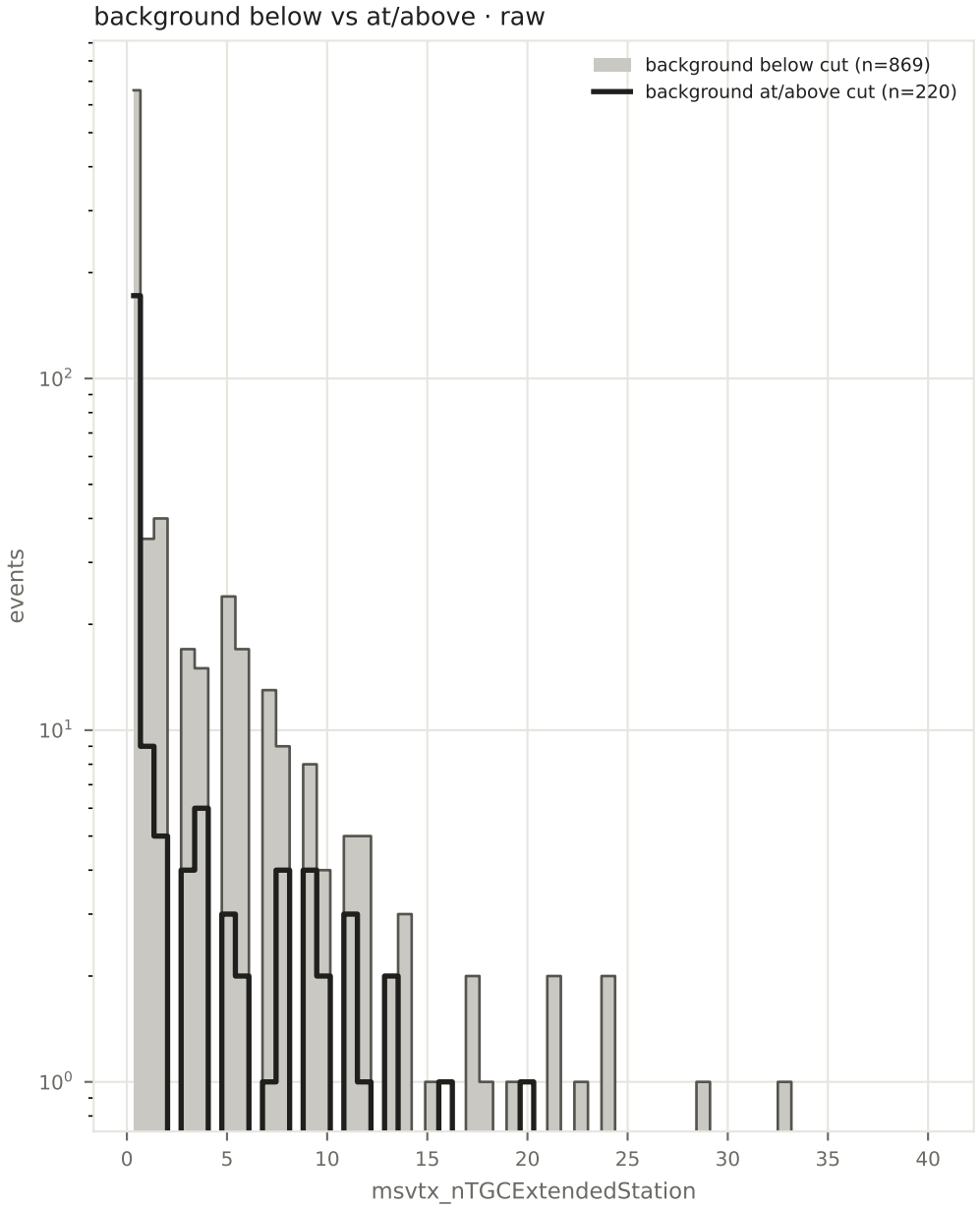


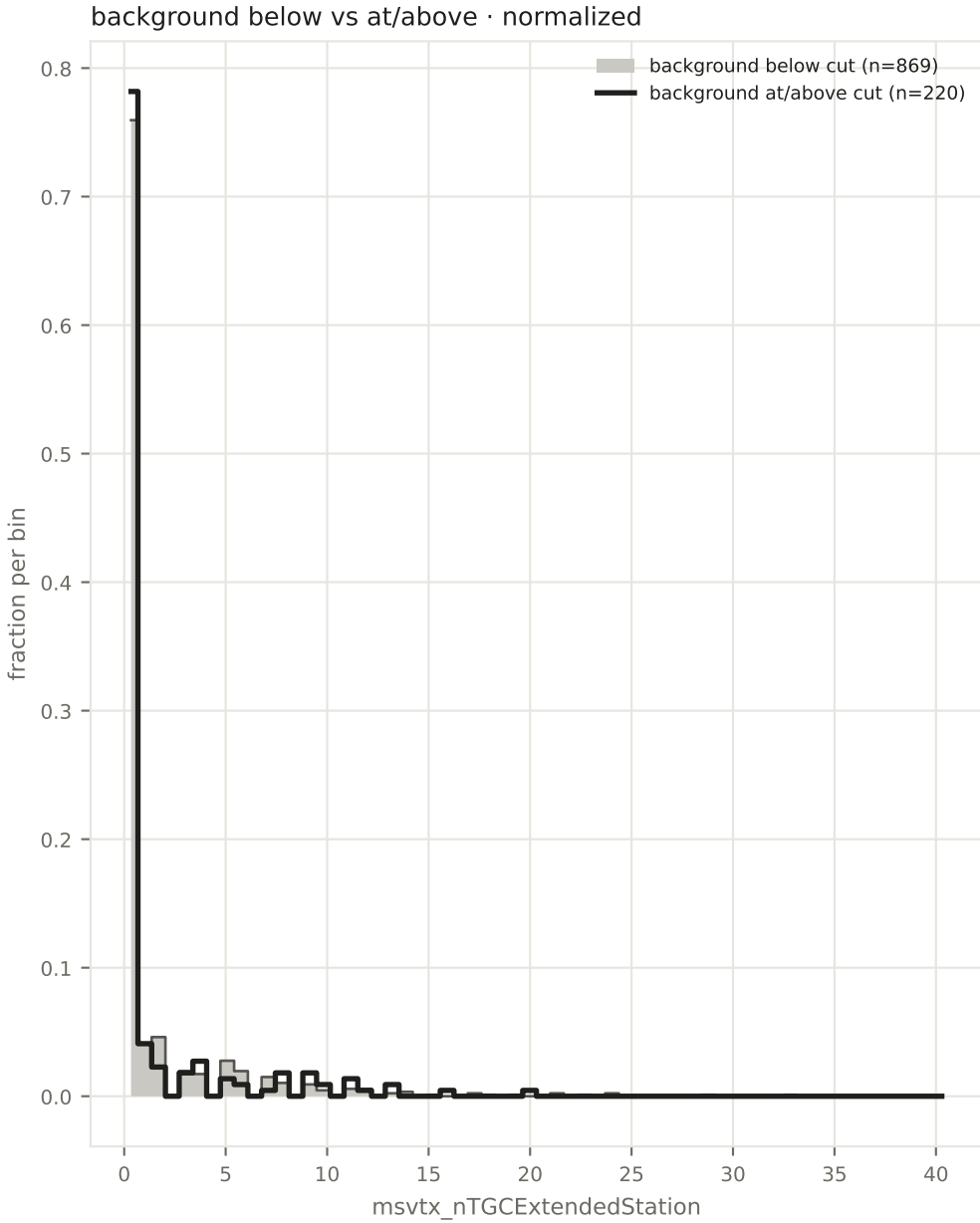
edge bins include under/overflow · NaN and -999 removed

msvtx_nTGCExtendedStation

region: BARREL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 6 / 169

D(below vs above) 0.027 n below = 869 n above = 220 large D = background shape changes across the cut



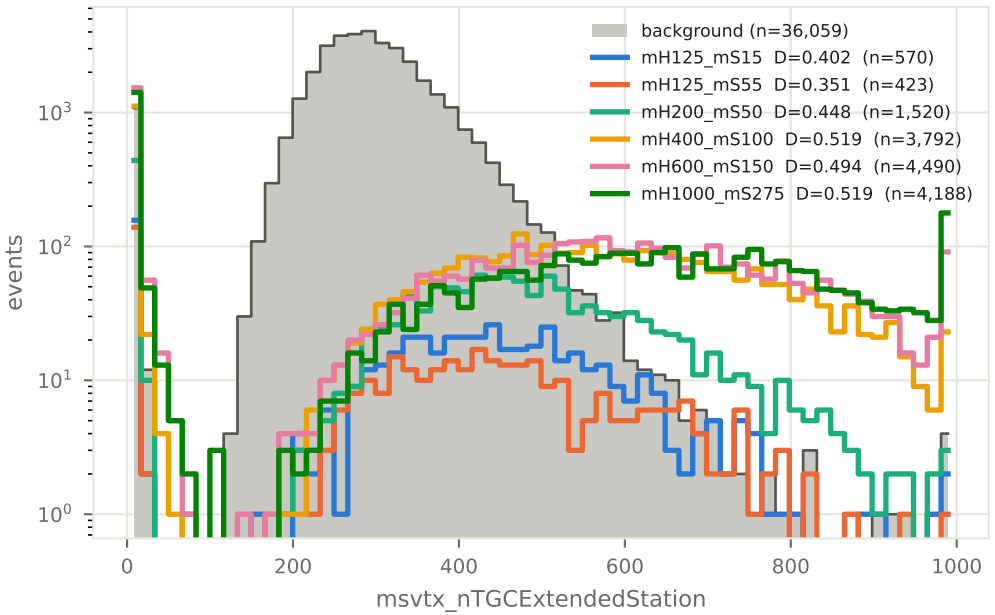


msvtx_nTGCExtendedStation

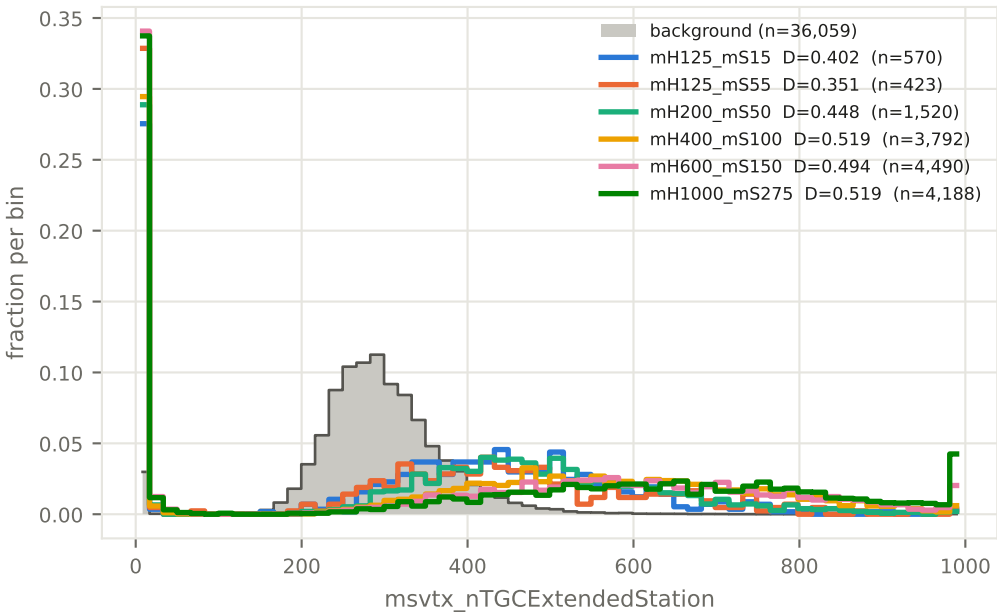
region: ALL · selection: all (every event) · rank 6 / 169

D_equal 0.448 D_pooled 0.490 min 0.351 (mH125_mS55) max 0.519 (mH1000_mS275) bkg n = 36,059 sig n = 14,983

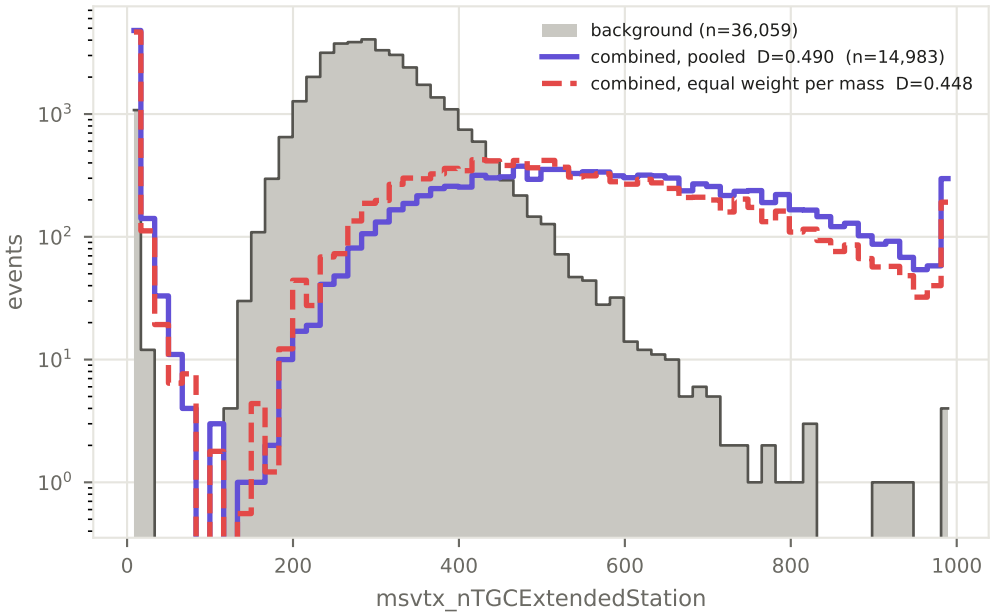
each mass point vs background · raw



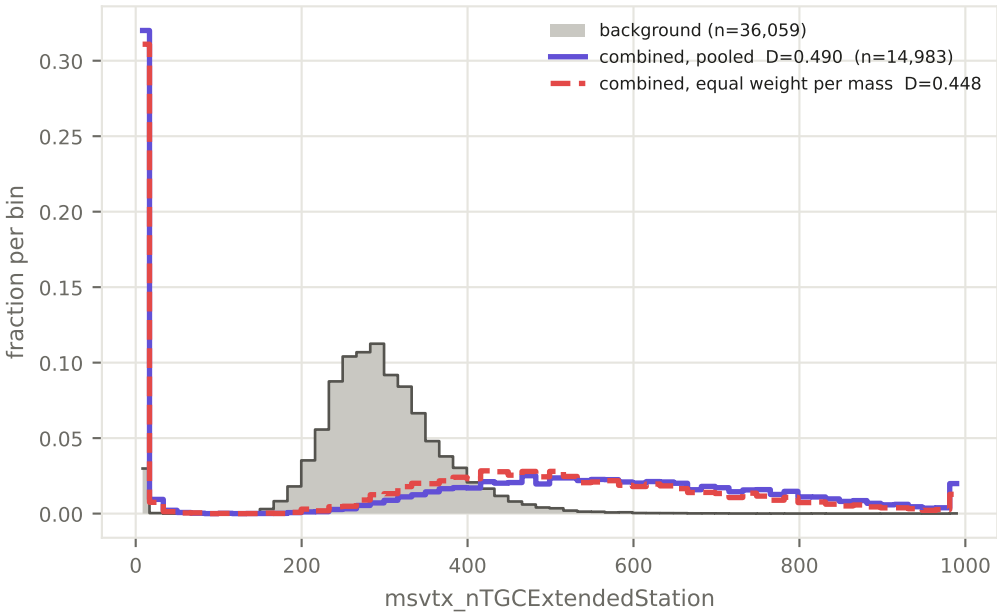
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



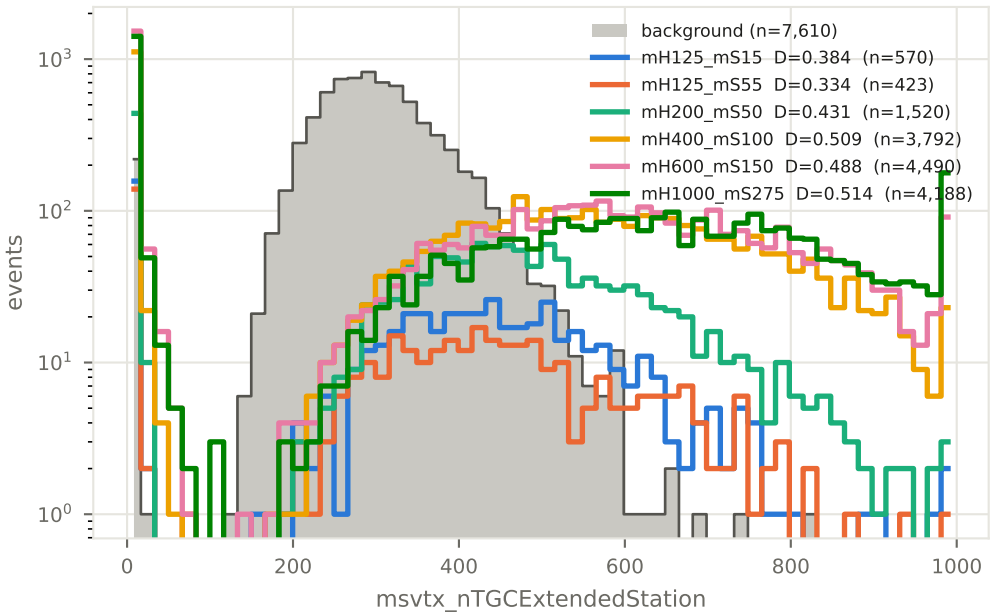
edge bins include under/overflow · NaN and -999 removed

msvtx_nTGCExtendedStation

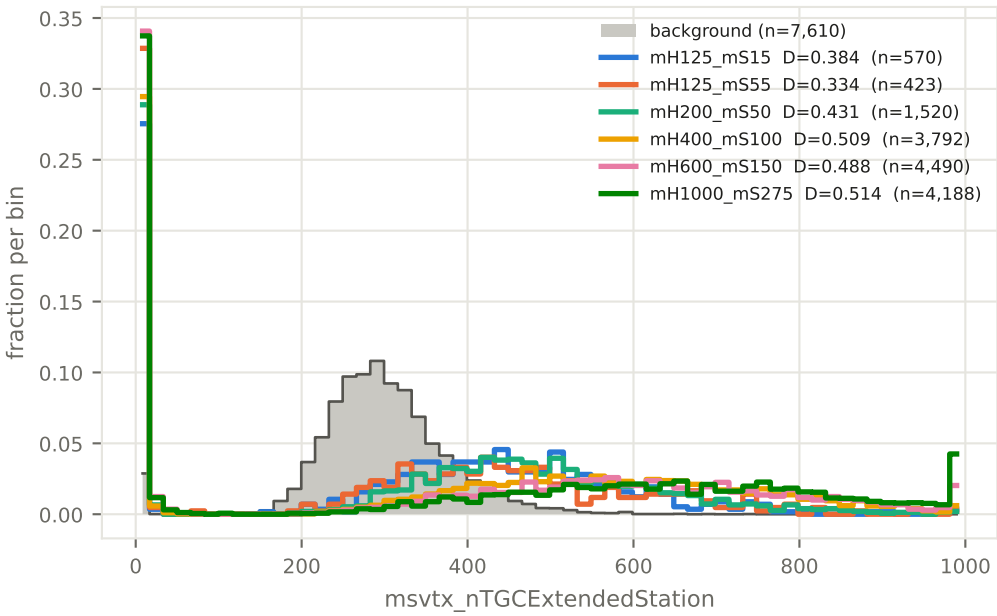
region: ALL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 6 / 169

D_equal 0.436 D_pooled 0.482 min 0.334 (mH125_mS55) max 0.514 (mH1000_mS275) bkg n = 7,610 sig n = 14,983

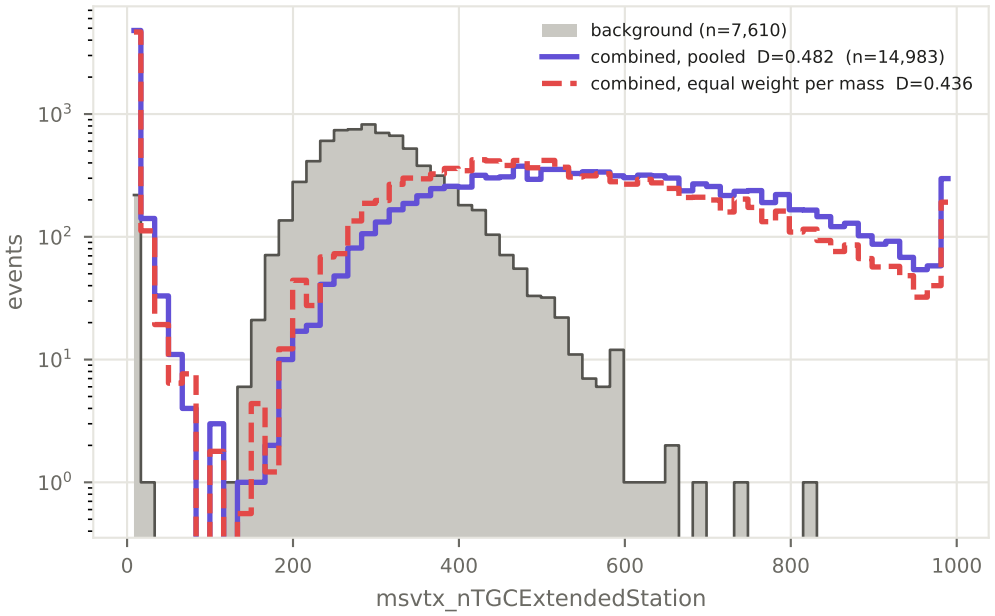
each mass point vs background · raw



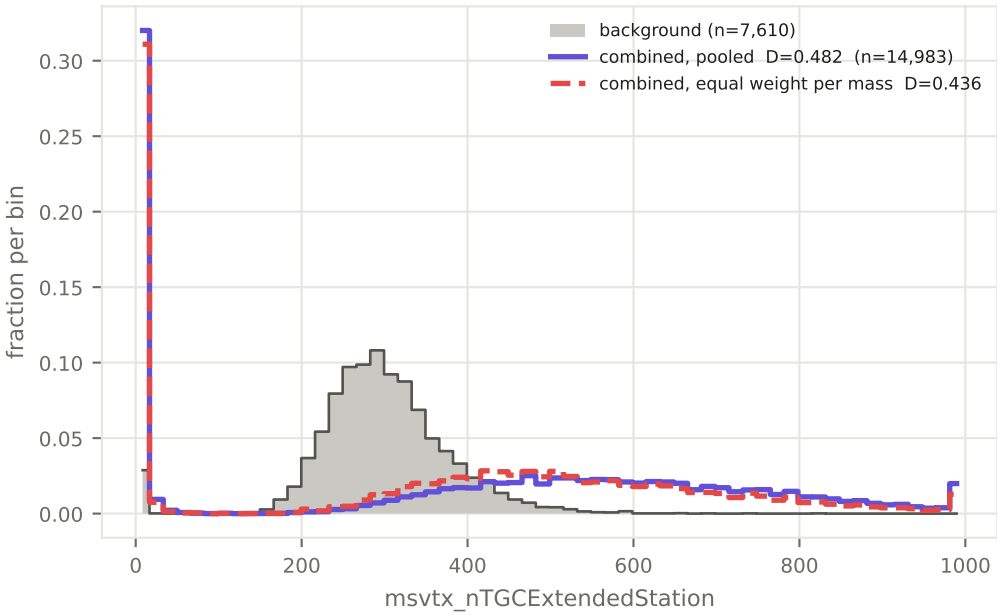
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



edge bins include under/overflow · NaN and -999 removed

msvtx_nTGCExtendedStation

region: ALL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 6 / 169

D(below vs above) 0.039 n below = 28,449 n above = 7,610 large D = background shape changes across the cut

